UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Introduction to Partial Differentiation

Complete study pack for class 001: functions of several variables, limits and continuity of a function of two variables, the definition and notation of partial derivatives, and second-order and mixed partial derivatives

Compiled from the PESU class 001 slide deck, its two AV-summary transcripts, the Unit 1 class-work sheet and past ESA papers. Generated 15 October 2026.

· How to use this pack

Class 001 has one deck and two recorded lectures: “Introduction to Partial Differentiation” (topics 1–2) and “Continuity and differentiability of a function of two independent variables” (topics 3–5). Topics 1–3 are the background (functions of two variables, limits, continuity) and are never asked by themselves in the ESA; topics 4–5 (computing partial derivatives and mixed partials) are used in every Unit 1 question. The geometrical interpretation and the problems on higher-order partials announced at the end of the second lecture are class 002, a separate pack. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a

Contents

1. 1 · Functions of two or more variables
2. 2 · Limit of a function of two variables
3. 3 · Continuity of a function of two variables
4. 4 · Partial derivatives: history, definition and notation
5. 5 · Second-order and mixed partial derivatives
6. Appendix A · Symbol glossary
7. Appendix B · Sources, coverage and gaps

1 Functions of two or more variables

Most quantities depend on more than one variable: the area of a rectangle on its length and breadth, the volume of a cylinder on r and h, a potential on x, y, z, t. A function of two variables $z = f(x, y)$ assigns to each ordered pair (x, y) in a domain D a unique real number. Its graph is a surface in three dimensions, so tools are used to plot it; this unit works with two variables, and everything extends to n variables.

Key terms and definitions

Function of two variables. A function of two variables denoted by $z = f(x, y)$ is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes on: $\{f(x, y) : (x, y) \in D\}$.
S001·8 · T001a 00:05:46

Independent and dependent variables. In $z = f(x, y)$, x and y are independent variables and z is the dependent variable (a bivariate function); with three independent variables one writes $w = f(x, y, z)$, and in general $f(x_1, x_2, \ldots, x_n)$. (transcript only) T001a 00:02:18 · T001a 00:02:54

Graphs and diagrams

$$z = \sin x \cos y$$

$$-\pi \le x,\ y \le \pi;\ \ -1 \le z \le 1$$

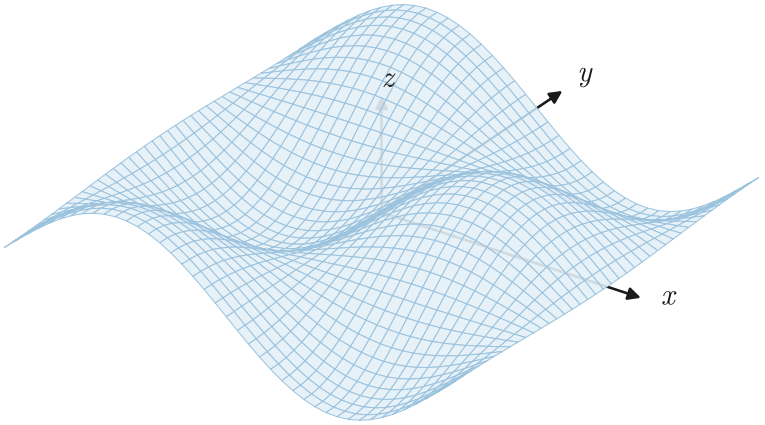


Figure G1.1. Graph of $z = \sin x \cos y$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A function of two variables is a surface over the xy-plane: above each point (x, y) the height is $f(x, y)$. $\sin x \cos y$ is an egg-box of hills and hollows between -1 and 1; the slide's Gnuplot picture shows the same surface.

$z = e^{1 - x^2 - y^2}$

top $(0, 0, e)$

green: level curves $z =$ const (circles)

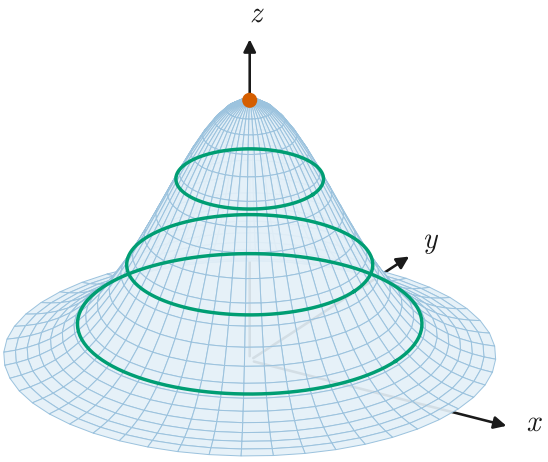


Figure G1.2. Graph of $z = e^{1-x^2-y^2}$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A single bell-shaped hill with its top $z = e$ above the origin, falling to 0 in every direction. The green level curves (drawn on the surface) are circles $x^2 + y^2 =$ constant.

Important points and why

1. Examples on the slides: area of a rectangle $A(l, b)$; volume of a cylinder $v(r, h)$; electric potential in a cathode-ray tube $V = f(x, y, z, t)$; propagation of an electromagnetic wave (magnetic field, electric field, wavelength, direction); weather (topography, pressure, humidity, temperature); spread of a pandemic (area, population, awareness, immunity, age and gender distribution). ESA ★ S001·4 · S001·5 · S001·6 · S001·7 · T001a 00:03:26
2. A single-variable function is enough to understand a concept, but to draw a conclusion about a real situation we almost always need several variables. ESA ★ TRANSCRIPT ONLY T001a 00:05:16
3. f maps an ordered pair to one real number: for $z = x^3y$, the pair $(2, 1)$ is mapped to 8. ESA ★ T001a 00:07:16
4. Graph of $z = f(x, y)$: a surface, which needs three axes; a function of three variables would need four dimensions, so the unit sticks to two variables. The slide shows $\sin x \cos y$ and $e^{1-x^2-y^2}$ (Figures G1.1–G1.2). ESA ★ S001·8 · T001a 00:08:35 · T001a 00:09:07

SHORT-NOTE RECAP

Quantities depending on several variables need functions of several variables (area $A(l,b)$, volume $v(r,h)$, potential $V(x,y,z,t)$, weather, pandemics). $z=f(x,y)$ assigns to each $(x,y)\in D$ a unique real number; D is the domain, the set of values the range. The graph is a surface.

2 Limit of a function of two variables

In one variable x can approach a only from the left or the right. In the plane (x, y) can approach (a, b) along infinitely many paths: lines, parabolas, ellipses. The limit is L only if every path gives L. Proving a limit exists is hard; proving it does not exist is easy: find two paths that give different values.

Key terms and definitions

Limit of a function of two variables. A function $f(x, y)$ is said to approach the limit L as (x, y) approaches (a, b), if the limit remains the same along any path of (x, y) approaching (a, b): $\lim_{(x,y)\to(a,b)} f(x, y) = L$. S001·9 · T001a 00:11:01

Formulas

E2.1 · Limit of f(x, y) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

gives	The value f approaches near (a, b).
applies	The same L along every path to (a, b).
note	For a polynomial (or any function continuous at the point) just substitute.

S001·9 · T001a 00:13:59

E2.2 · Two-path test (limit does not exist) ESA ★

$$\lim_{\text{path 1}} f \neq \lim_{\text{path 2}} f \implies \lim_{(x,y)\to(a,b)} f(x, y) \text{ does not exist}$$

gives	A proof that the limit does not exist.
applies	Substitution gives $\frac{0}{0}$.
note	Lines $y = mx$ through the origin are the usual first try: if the answer depends on m, the limit does not exist.

S001·10 · T001a 00:17:42

Graphs and diagrams

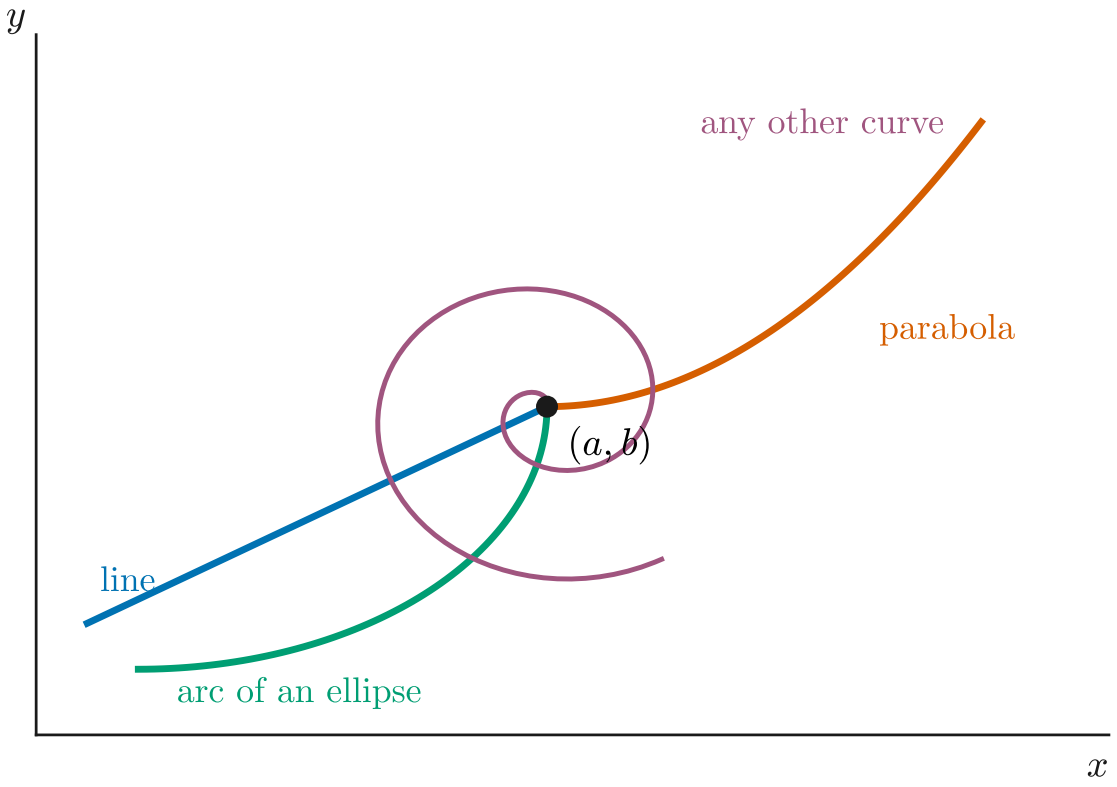


Figure G2.1. Paths along which (x, y) can approach (a, b). ESA ★

S001·9 · T001a 00:11:52 · T001a 00:12:19

How to read it. In one variable x reaches a only from the left or the right. In the plane, (x, y) can reach (a, b) along a line, a parabola, an arc of an ellipse, or any other path. The limit exists only if all of them give the same value.

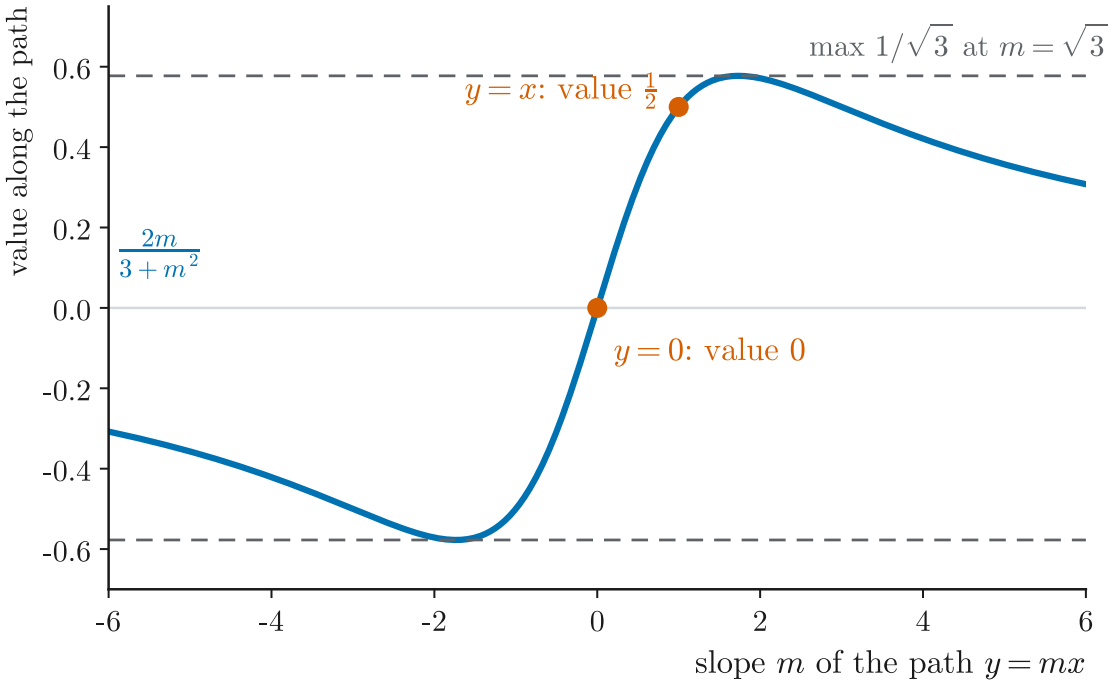
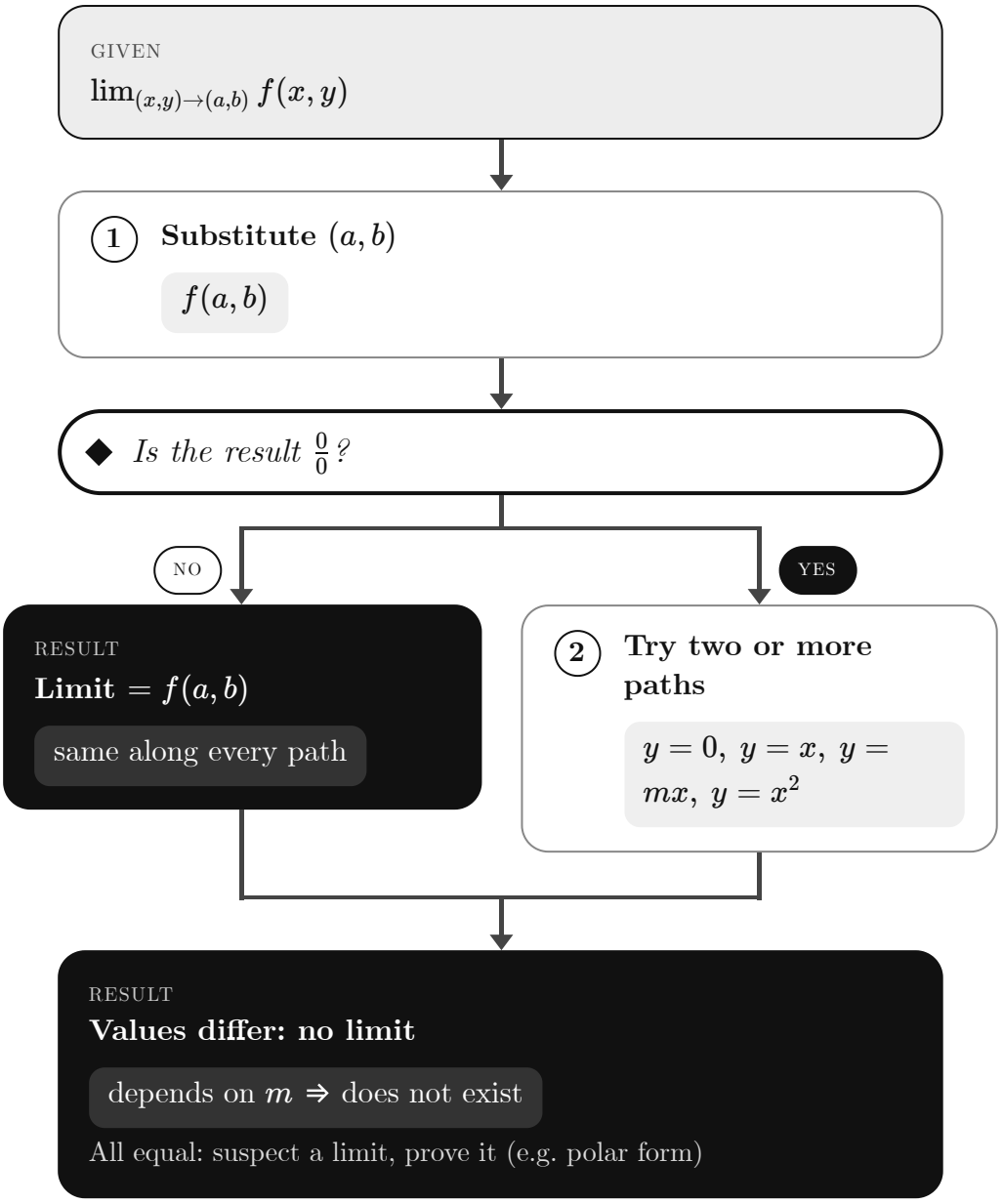


Figure G2.2. $\frac{2xy}{3x^2+y^2}$ along the line $y=mx$. ESA ★★

ADDED FOR CLARITY S001·10 · T001a 00:17:24

How to read it. The value of $\frac{2xy}{3x^2+y^2}$ on the line $y=mx$ is the constant $\frac{2m}{3+m^2}$, plotted against the slope m. $m=0$ gives 0, $m=1$ gives $\frac{1}{2}$, and the largest value is $\frac{1}{\sqrt{3}}$ at $m=\sqrt{3}$. Because the value changes with the path, the limit at the origin does not exist.

Flowcharts



Remark. Agreement along a few paths is only evidence; disagreement is proof.

Exam trap. Read the point carefully: slide $(2,1)$ gives -8, the lecture's $(2,-1)$ gives -6.

Figure F2.1. Finding (or disproving) a limit at a point ESA ★★

S001·9 · S001·10 · T001a 00:13:59 · T001a 00:17:42

One variable and two variables

	$y = f(x)$	$z = f(x, y)$
Domain	Real numbers	Ordered pairs (x, y)
Graph	A curve in the plane	A surface in space
Approach to the point	From the left or the right	Along infinitely many paths
Limit exists if	Left and right limits agree	Every path gives the same value
Continuous if	$\lim_{x \to c} f = f(c)$	$\lim_{(x,y) \to (a,b)} f = f(a, b)$
Derivative	$\frac{dy}{dx}$	$\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ (others constant)

S001·9 · S001·11 · T001a 00:09:54 · T001a 00:11:52

Important points and why

1. One variable: $x \to a$ from the left or the right only. Two variables: $(x, y) \to (a, b)$ along any of infinitely many paths (Figure G2.1). ESA ★ S001·9 · T001a 00:11:32 · T001a 00:12:34
2. If substitution gives a number (a polynomial, no $\frac{0}{0}$), that number is the limit, whatever the path. ESA ★ T001a 00:13:59
3. It is hard to show that a limit exists but easy to show it does not: two paths with different values are enough. ESA ★★ S001·10 · T001a 00:17:42
4. Agreement along a few paths proves nothing: for $\frac{2xy}{3x^2+y^2}$, the x-axis and the parabola $y = x^2$ both give 0, but $y = x$ gives $\frac{1}{2}$. ESA ★★ T001a 00:16:48 · T001a 00:17:24

Why can one path never prove that a limit exists?
There are infinitely many paths, and a function can agree on any finite number of them and still differ on another: $\frac{2xy}{3x^2+y^2}$ gives 0 on the x-axis and on $y = x^2$ but $\frac{1}{2}$ on $y = x$. To prove existence one needs an estimate for all paths at once (e.g. polar coordinates: $\frac{3x^2y}{x^2+y^2} = 3r\cos^2\theta\sin\theta \to 0$).
ADDED FOR CLARITY T001a 00:17:42 · S001·12

Exam trap. Same value along two or three paths does not prove that the limit exists; different values along two paths does prove it does not. T001a 00:16:48 · T001a 00:17:42

Exam trap. Slide 001·9 uses the point $(2, 1)$ (limit -8); the lecture uses $(2, -1)$ (limit -6). Read the point in the question. ADDED FOR CLARITY S001·9 · T001a 00:12:59

Worked examples

W2.1 · Limit of a polynomial (slide 001·9) ESA ★

GIVEN Evaluate $\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$; the lecture also does the point $(2, -1)$.

STEP 1 A polynomial is continuous everywhere, so substitute.

STEP 2 At $(2,1)$: $4 - 4 + 3 - 8 + 3 - 6 = -8$.

STEP 3 At $(2,-1)$: $4 + 4 + 3 - 8 - 3 - 6 = -6$.

ANSWER -8 at $(2,1)$; -6 at $(2,-1)$

CHECK sympy agrees for both points. The value is reached along every path.

S001·9 · T001a 00:12:59 · T001a 00:13:47

W2.2 · Show that a limit does not exist (slide 001·10) ESA ★★

GIVEN Show that $\lim_{(x,y)\to(0,0)} \dfrac{2xy}{3x^2 + y^2}$ does not exist.

STEP 1 Substitution gives $\frac{0}{0}$: indeterminate.

STEP 2 Along $y = 0$ (x-axis): $\frac{0}{3x^2} = 0 \to 0$.

STEP 3 Along $y = x^2$: $\frac{2x^3}{3x^2+x^4} = \frac{2x}{3+x^2} \to 0$.

STEP 4 Along $y = x$: $\frac{2x^2}{4x^2} = \frac{1}{2}$.

STEP 5 General line $y = mx$: $\frac{2m}{3+m^2}$, which depends on m (Figure G2.2).

ANSWER Different paths give 0 and $\frac{1}{2}$: the limit does not exist

CHECK sympy: $\frac{2m}{m^2+3}$ along $y = mx$.

S001·10 · T001a 00:14:20 · T001a 00:15:04 · T001a 00:17:03

SHORT-NOTE RECAP

$\lim_{(x,y)\to(a,b)} f = L$ if every path to (a,b) gives L. Polynomial: substitute. $\frac{0}{0}$: try paths ($y = 0$, $y = x$, $y = mx$, $y = x^2$); two different values ⇒ no limit. $\frac{2xy}{3x^2+y^2}$: 0 on $y = 0$, $\frac{1}{2}$ on $y = x$.

3 Continuity of a function of two variables

f is continuous at (a,b) when $f(a,b)$ is defined, the limit exists, and $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$, as in one variable. Geometrically the surface has no hole or break there. A function can fail to be continuous even though the limit exists ($\frac{3x^2y}{x^2+y^2}$ at the origin), or because the limit does not exist ($\frac{2xy}{3x^2+y^2}$).

Key terms and definitions

Continuity at a point. $f(x,y)$ is continuous at (a,b) if $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$: $f(a,b)$ must be defined, the limit must exist, and the two must be equal. S001·11 · T001b 00:00:38 · T001a 00:18:46

> *"A function is continuous at a point if the graph doesn't have any holes or breaks at that point. Graphing the function is helpful. If a function is not continuous at (a,b), this does not mean the limit does not exist. We will need to consider the approaching (a,b) from different paths."* S001·11

Formulas

E3.1 · Continuity at (a, b) ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

gives	The test for continuity.
applies	Three conditions: $f(a,b)$ defined, limit exists, both equal.
note	One variable: $\lim_{x\to c} f(x) = f(c)$.

S001·11 · T001b 00:00:38

Graphs and diagrams

$$z = \frac{3x^2 y}{x^2 + y^2}$$

limit 0 at the origin; $f(0,0)$ undefined

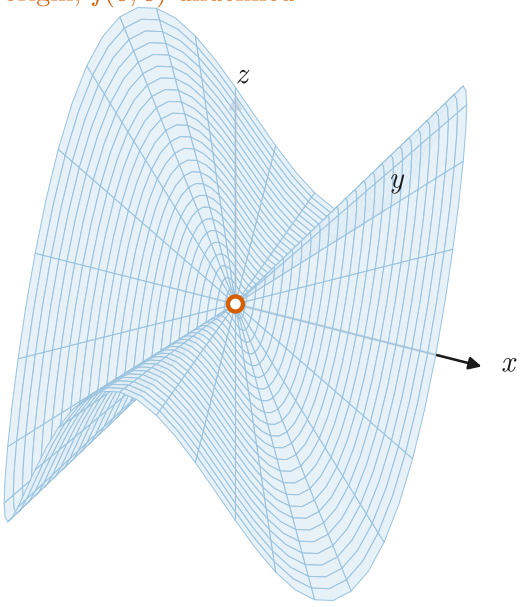


Figure G3.1. $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists. ESA ★

S001·12 · T001b 00:02:41

How to read it. The surface folds near the origin but every direction leads to height 0 there (vermilion point). It is missing only the single point $(0,0)$, where f is undefined: a removable discontinuity.

$$z = \frac{2xy}{3x^2 + y^2}$$

ray $y = x$: height $\frac{1}{2}$

x-axis: height 0

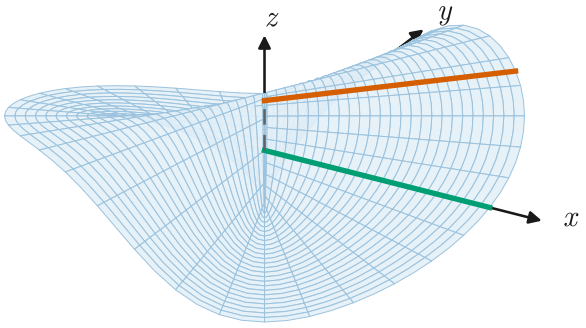
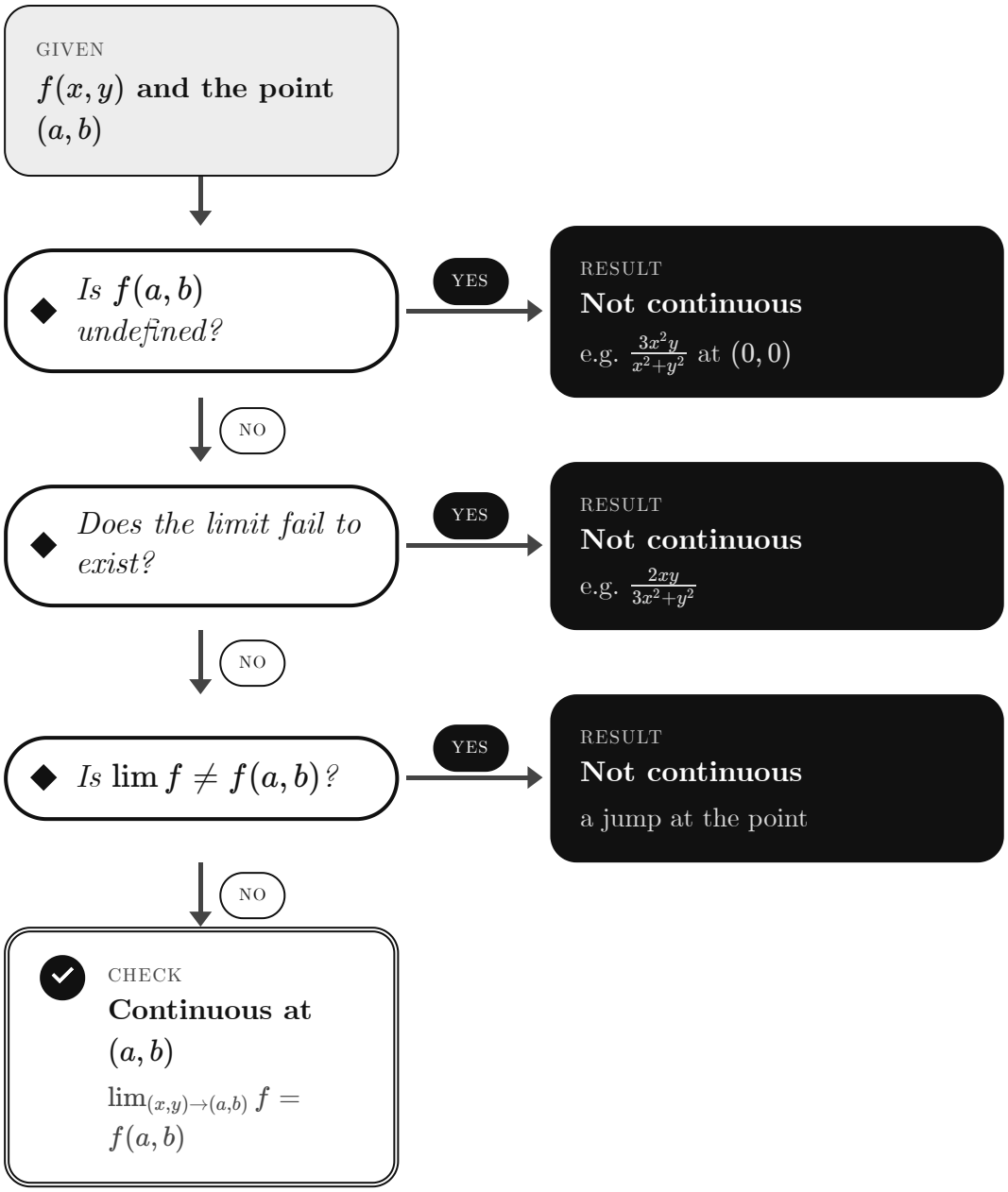


Figure G3.2. $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist. ESA ★

S001·13 · T001b 00:06:13

How to read it. Each ray from the origin keeps a constant height $\frac{2m}{3+m^2}$, so the surface is a twisted fan of straight lines meeting at the z-axis at different heights (vermilion: the ray $y = x$ at height $\frac{1}{2}$; blue: the x-axis at height 0). There is no single value at the origin.

Flowcharts



Remark. Only the first rung failing with the limit existing is a removable discontinuity: define $f(a,b)$ as the limit.

Figure F3.1. Is f continuous at (a, b)? ESA ★

S001·11 · T001a 00:18:46 · T001b 00:03:17

Important points and why

1. One variable: f is continuous at c if (i) $\lim_{x\to c} f(x)$ exists, (ii) $f(c)$ exists, (iii) they are equal. The same three conditions define continuity for $f(x,y)$. ESA ★ T001a 00:18:46 · S001·11

2. $\frac{3x^2y}{x^2+y^2}$: the limit at the origin exists (it is 0) but $f(0,0)$ is $\frac{0}{0}$, so f is not continuous there. $\frac{2xy}{3x^2+y^2}$: the limit does not exist and $f(0,0)$ is undefined (Figures G3.1–G3.2). ESA ★ S001·12 · S001·13 · T001b 00:04:14 · T001b 00:05:44
3. To decide continuity in higher dimensions, graph the function: a hole, fold or sudden drop in the surface shows a discontinuity. ESA ★ S001·11 · T001b 00:06:45 · T001a 00:18:07

Why is $\frac{3x^2y}{x^2+y^2}$ not continuous at the origin even though its limit is 0?
Continuity needs $f(0,0)$ to be defined and equal to the limit. Here $f(0,0)$ is $\frac{0}{0}$, so the third condition cannot be checked. Defining $f(0,0)=0$ would make it continuous.
ADDED FOR CLARITY S001·12 · T001b 00:04:14

Exam trap. "Not continuous" does not mean "the limit does not exist" (slide 001·11): $\frac{3x^2y}{x^2+y^2}$ has limit 0 at the origin. S001·11 · T001b 00:02:10

Exam trap. Continuous is not the same as smooth: $z=|x|+|y|$ is continuous everywhere but has a corner at the origin. The lecture's "continuous if the surface is smooth" is loose.
ADDED FOR CLARITY T001b 00:06:26

Worked examples

W3.1 · Limit exists, function not continuous (slide 001·12) ESA ★

GIVEN Discuss the continuity of $f(x,y)=\dfrac{3x^2y}{x^2+y^2}$ at the origin.

STEP 1 $f(0,0)=\frac{0}{0}$ is not defined, so condition (ii) fails: f is not continuous at $(0,0)$.

STEP 2 The limit exists: in polar form $x=r\cos\theta$, $y=r\sin\theta$, $f=3r\cos^2\theta\sin\theta$, and $|f|\le 3r\to 0$ for every path (added for clarity).

STEP 3 So $\lim_{(x,y)\to(0,0)} f=0$.

ANSWER Not continuous at $(0,0)$; the limit is 0 (removable: set $f(0,0)=0$)

CHECK sympy: polar form $3r\sin\theta\cos^2\theta$.

S001·12 · T001b 00:02:41 · T001b 00:04:27

W3.2 · Limit does not exist, function not continuous (slide 001·13) ESA ★

GIVEN Discuss the continuity of $f(x,y)=\dfrac{2xy}{3x^2+y^2}$ at the origin.

STEP 1 The limit does not exist (W2.2).

STEP 2 $f(0,0)$ is also undefined.

ANSWER Not continuous at $(0,0)$; no choice of $f(0,0)$ can make it continuous

CHECK Figure G3.2 shows the jump along different directions.

S001·13 · T001b 00:05:05 · T001b 00:05:44

SHORT-NOTE RECAP

Continuous at (a,b): $f(a,b)$ defined, limit exists, $\lim f = f(a,b)$; no hole or break in the surface.
Not continuous ≠ no limit: $\frac{3x^2y}{x^2+y^2}$ has limit 0 but $f(0,0)$ undefined.

4 Partial derivatives: history, definition and notation

A partial derivative is the derivative with respect to one variable with the others held constant. The idea is already in Leibniz's product rule; the symbol ∂ came from Legendre (1786) and was reintroduced by Jacobi (1841). $\frac{\partial z}{\partial x} = z_x = f_x$ is defined by a limit, and every rule of ordinary differentiation (product, quotient, chain) still holds.

Key terms and definitions

Partial derivative. A partial derivative of a function of several variables is the derivative with respect to one of those variables, with the others held constant. S001·15 · T001b 00:12:52

Formulas

E4.1 · Partial derivatives from the definition ESA ★★

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$
$$\frac{\partial z}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

gives	z_x (y held constant) and z_y (x held constant).
applies	When the limits exist.
note	Only one variable is moved by h; the other stays fixed.

S001·15

E4.2 · Notation for first-order partials ESA ★★

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y$$

gives	The equivalent symbols.
applies	$z = f(x, y)$.
note	The symbol is ∂ (“partial”, often read “dou”), not δ or d.

S001·15 · T001b 00:11:41 · T001b 00:13:05

Flowcharts

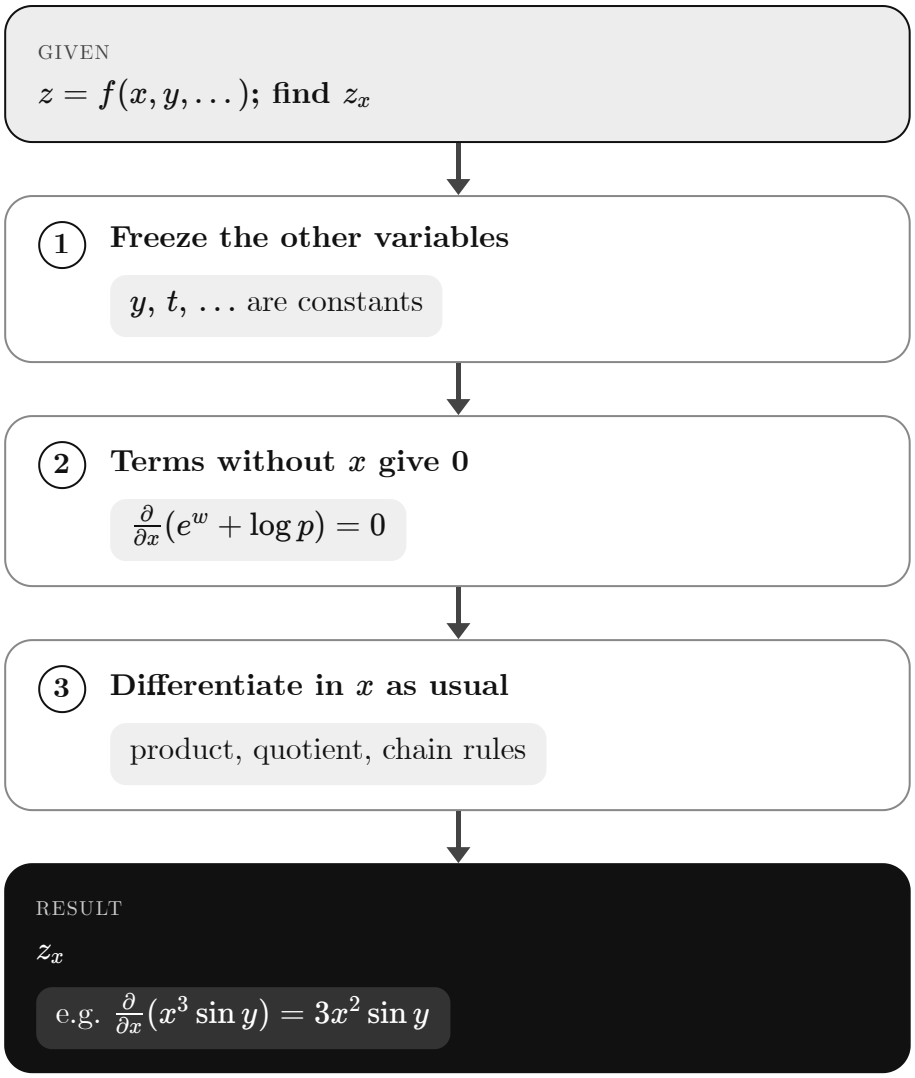


Figure F4.1. Computing a partial derivative ESA ★★

S001·15 · T001b 00:15:08 · T001b 00:13:57

Important points and why

1. The product rule is already partial differentiation: in $\frac{d}{dx}(x^3 \sin x) = x^3 \cos x + 3x^2 \sin x$, each term treats the other factor as a constant k. Leibniz: $d(uv) = u\,dv + v\,du$. ESA ★ S001·14 · T001b 00:07:56 · T001b 00:10:18
2. History: Leibniz first used partial differentiation (product rule); Adrien-Marie Legendre created the modern symbol ∂ in 1786 but later abandoned it; Carl Gustav Jacob Jacobi reintroduced it in 1841. ESA ★ S001·14 · T001b 00:11:23
3. To find z_x, look only at the terms containing x; everything else is a constant, however many other variables there are. ESA ★★ T001b 00:15:08 · T001b 00:15:48
4. A function of x, y, t has three first partials z_x, z_y, z_t; for z_x treat y and t as constants. ESA ★ T001b 00:13:20
5. All rules of ordinary differentiation hold for partial derivatives: product, quotient, chain rule, implicit and composite functions. ESA ★★ T001b 00:13:57 · T001b 00:14:16

Why does every rule of ordinary differentiation still work?
With the other variables frozen, $f(x, y)$ is an ordinary function of x alone, and $\frac{\partial f}{\partial x}$ is its ordinary derivative (E4.1 is the usual limit with y fixed).

T001b 00:13:57 · S001·15

Exam trap. The partial symbol is ∂, not δ (the lecture stresses this) and not d. T001b 00:11:41

Worked examples

W4.1 · A partial derivative from the definition — ESA ★

given Use E4.1 to find f_x for $f(x, y) = x^2 y$.

step 1 $\frac{f(x+h,y)-f(x,y)}{h} = \frac{(x+h)^2 y - x^2 y}{h} = \frac{(2xh+h^2)y}{h} = (2x + h)y.$

step 2 Let $h \to 0$: $2xy$.

answer $\boxed{f_x = 2xy}$

check Same as differentiating x^2 with y constant; sympy agrees.

added for clarity S001·15

W4.2 · Partials of a function of several variables (lecture) — ESA ★★

given $z = x^3 \sin y + e^w + \log p$ (a function of x, y, w, p). Find z_x, z_p, z_y and z_{xy}.

step 1 z_x: only $x^3 \sin y$ contains x; $\sin y$ is a constant: $3x^2 \sin y$.

step 2 z_p: only $\log p$ contains p: $\frac{1}{p}$.

step 3 z_y: only $\sin y$ varies: $x^3 \cos y$.

step 4 z_{xy}: differentiate $z_x = 3x^2 \sin y$ with respect to y: $3x^2 \cos y$.

answer $\boxed{z_x = 3x^2 \sin y,\ z_p = \dfrac{1}{p},\ z_y = x^3 \cos y,\ z_{xy} = 3x^2 \cos y}$

check sympy agrees; $z_{yx} = \frac{\partial}{\partial x}(x^3 \cos y) = 3x^2 \cos y$ too (Clairaut).

T001b 00:14:40 · T001b 00:15:48 · T001b 00:16:26 · T001b 00:16:44

short-note recap

Partial derivative: derivative in one variable, others constant. $z_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$; notation $\frac{\partial z}{\partial x} = z_x = f_x$. Leibniz (product rule), Legendre 1786, Jacobi 1841. All differentiation rules hold.

5 Second-order and mixed partial derivatives

Differentiating again gives z_{xx}, z_{yy} and the mixed partials z_{xy}, z_{yx}. Clairaut's theorem: if the second derivatives are continuous, the order of differentiation does not matter, $z_{xy} = z_{yx}$. Subscripts are read in the order of differentiation; in the ∂ notation the order is read from right to left.

Key terms and definitions

Mixed (crossed) partial derivatives. $z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$ and $z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$: differentiate with respect to one variable, then the other. S001·16 · T001b 00:18:24

> *"Note 1: The crossed or mixed partial derivatives are in general equal (Clairaut's theorem): $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$. That is, the order of differentiation is immaterial if the second order derivatives involved are continuous."* S001·17 · T001b 00:19:57

> *"Note 2: In the subscript notation, the subscripts are written in same order in which the differentiation is carried out, while in the ∂ notation the order is opposite. For example, $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = f_{xy}$."* S001·17

Formulas

E5.1 · Second-order partials ESA ★★

$$z_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} = f_{xx}, \qquad z_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} = f_{yy}$$

GIVES	Second derivatives in one variable.
APPLIES	Differentiate z_x again in x (or z_y in y).
NOTE	Same rules as ordinary second derivatives.

S001·16 · T001b 00:17:04

E5.2 · Mixed partials ESA ★★

$$z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y\,\partial x} = f_{xy}, \qquad z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x\,\partial y} = f_{yx}$$

GIVES	The two crossed derivatives.
APPLIES	Differentiate in one variable, then the other.
NOTE	Subscripts: left to right. ∂ notation: right to left.

S001·16 · S001·17 · T001b 00:18:41

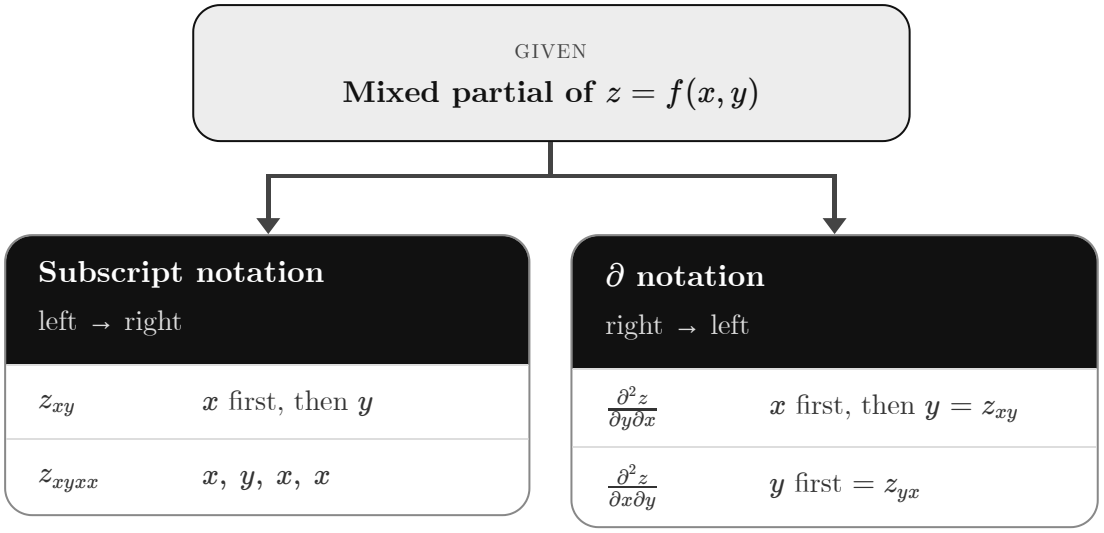
E5.3 · Clairaut's theorem ESA ★★★

$$f_{xy} = f_{yx} \qquad \left(\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}\right)$$

GIVES	The order of differentiation does not matter.
APPLIES	f_{xy} and f_{yx} continuous near the point.
NOTE	Called "a very important theorem" in the lecture; it is why Taylor's series has $2hkf_{xy}$.

S001·17 · T001b 00:19:57 · T001b 00:20:10

Flowcharts



Remark. By Clairaut's theorem the order rarely changes the answer, but the notation is still examined.

Figure F5.1. Reading the order of a mixed partial ESA ★★

S001·17 · T001b 00:18:51

Important points and why

1. Read z_{xyxx} left to right: $z \to z_x \to z_{xy} \to z_{xyx} \to z_{xyxx}$ (differentiate in x, then y, then x, then x). ESA ★ T001b 00:18:51 · T001b 00:19:21
2. Clairaut's theorem is "a very important theorem": whenever the second partials are continuous (true for all examination functions away from bad points), compute whichever mixed partial is easier. ESA ★★★ S001·17 · T001b 00:20:10
3. Verifying $f_{xy} = f_{yx}$ is a standard class-work question: x^y (CW P2) and $x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$ (CW P8, W5.3). ESA ★★ CW U1 P2 · CW U1 P8

Exam trap. $\frac{\partial^2 z}{\partial y\,\partial x}$ means x first, then y: it is z_{xy}, not z_{yx}. S001·17

Exam trap. Clairaut needs continuous second partials: for $f = \frac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0)=0$, one gets $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = 1$. ADDED FOR CLARITY S001·17

Worked examples

W5.1 · Second-order partials of a function of three variables

ESA ★★

GIVEN If $f(x,y,z) = e^{x^2+y^2+z^2}$, find f_{xx}, f_{yy}, f_{zz} at $(-1,1,-1)$.

STEP 1 $f_x = 2xe^{x^2+y^2+z^2}$.

STEP 2 $f_{xx} = (2+4x^2)e^{x^2+y^2+z^2}$ (product rule).

STEP 3 At $(-1,1,-1)$: $x^2+y^2+z^2 = 3$, so $f_{xx} = 6e^3$; by symmetry $f_{yy} = (2+4y^2)e^3 = 6e^3$, $f_{zz} = 6e^3$.

ANSWER $\boxed{f_{xx} = f_{yy} = f_{zz} = 6e^3}$

CHECK CW answer agrees; sympy agrees.

CW U1 P4

W5.2 · Find n so that v satisfies an equation

ESA ★★

GIVEN Find n so that $v = r^n(3\cos^2\theta - 1)$ satisfies $\dfrac{\partial}{\partial r}\left(r^2\dfrac{\partial v}{\partial r}\right) + \dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial v}{\partial\theta}\right) = 0$.

STEP 1 $v_r = nr^{n-1}(3\cos^2\theta - 1)$, so $\frac{\partial}{\partial r}(r^2 v_r) = n(n+1)r^n(3\cos^2\theta - 1)$.

STEP 2 $v_\theta = -6r^n\cos\theta\sin\theta$, so $\sin\theta\, v_\theta = -6r^n\cos\theta\sin^2\theta$.

STEP 3 $\frac{\partial}{\partial\theta}(\cdot) = -6r^n(2\sin\theta\cos^2\theta - \sin^3\theta)$; divide by $\sin\theta$: $-6r^n(3\cos^2\theta - 1)$.

STEP 4 Sum: $[n(n+1) - 6]\, r^n(3\cos^2\theta - 1) = 0 \Rightarrow n^2 + n - 6 = 0$.

ANSWER $\boxed{n = 2 \text{ or } n = -3}$

CHECK CW answer $(-3,\ 2)$ agrees; sympy solves to $[-3, 2]$. Same pattern as ESA Jan-24 s4 1c (worked in the class 002 pack).

CW U1 P7 · ESA Jan-24 s4 1c

W5.3 · Verify that the mixed partials are equal ESA ★★

GIVEN If $u = x^2 \tan^{-1}\dfrac{y}{x} - y^2 \tan^{-1}\dfrac{x}{y}$, show that $\dfrac{\partial^2 u}{\partial x \partial y} = \dfrac{\partial^2 u}{\partial y \partial x} = \dfrac{x^2 - y^2}{x^2 + y^2}$.

STEP 1 $u_x = 2x \tan^{-1}\frac{y}{x} + x^2 \cdot \frac{-y}{x^2+y^2} - y^2 \cdot \frac{y}{x^2+y^2} = 2x \tan^{-1}\frac{y}{x} - y$.

STEP 2 $u_{xy} = \frac{\partial}{\partial y}(u_x) = \frac{2x^2}{x^2+y^2} - 1 = \frac{x^2-y^2}{x^2+y^2}$.

STEP 3 Similarly $u_y = x - 2y \tan^{-1}\frac{x}{y}$, and $u_{yx} = 1 - \frac{2y^2}{x^2+y^2} = \frac{x^2-y^2}{x^2+y^2}$.

ANSWER $\boxed{u_{xy} = u_{yx} = \dfrac{x^2 - y^2}{x^2 + y^2}}$

CHECK sympy: both simplify to $\frac{x^2-y^2}{x^2+y^2}$.

CW U1 P8

SHORT-NOTE RECAP

z_{xx}, z_{yy}, $z_{xy} = \frac{\partial}{\partial y}(z_x) = \frac{\partial^2 z}{\partial y \partial x}$, z_{yx}. Clairaut: $z_{xy} = z_{yx}$ when the second partials are continuous. Subscripts left to right, ∂ right to left.

A Symbol glossary

Symbol	Name	Meaning	Appears in
$z = f(x, y)$	Function of two variables	x, y independent, z dependent	T1
D	Domain	Set of ordered pairs where f is defined	T1
L	Limit	Common value along every path	E2.1
(a, b)	Point of approach	Where the limit or continuity is tested	E2.1, E3.1
m	Slope of a path	Path $y = mx$ in the two-path test	E2.2, W2.2
∂	Partial (“dou”)	Partial derivative sign; not δ, not d	E4.1–E5.3
h	Increment	Small change in one variable in the definition	E4.1
z_x, f_x	First partial in x	$\frac{\partial z}{\partial x}$ with y constant	E4.2
z_{xx}, z_{yy}	Second partials	$\frac{\partial^2 z}{\partial x^2}$, $\frac{\partial^2 z}{\partial y^2}$	E5.1
z_{xy}	Mixed partial	x first, then y; $= \frac{\partial^2 z}{\partial y \partial x}$	E5.2, E5.3

B Sources, coverage and gaps

Source	Path	Use
Deck 001 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.*	18 slides; content on 3–17
Transcript 001a (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.*	19.7 min: motivation, definition, limits
Transcript 001b (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.*	21.1 min: continuity, history, definition, mixed partials
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 1–8 (with answers)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	No direct question; two partial-derivative computations (class 002)
Videos 001a, 001b (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_*.mp4	Not opened: every slide is readable and the board work is read out in the transcripts

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No question asks for a limit, continuity or the definition of a partial derivative. Computing partial derivatives is asked directly twice: Dec 2023 Q1.a (third-order partials of $e^x \log y + \cos y \log x$) and Jan 2024 set 4 Q1.c (find n for $\theta = t^n e^{-r^2/4t}$). Both are worked in the class 002 pack; this pack works the class-work problems of the same type (CW P4, P7, P8). Every other Unit 1 question (chain rule, Euler, Taylor, maxima and minima, Lagrange) uses the rules of this session.

Known gaps, conflicts and corrections

1. Slide 001·9 asks for $\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$, which is -8. The lecture (T001a 00:12:59) works the point $(2, -1)$ instead and gets -6. Both values are correct for their points (sympy); W2.1 gives both.
2. Slide 001·12's text is about $f = \frac{3x^2 y}{x^2+y^2}$ (whose limit at the origin is 0), but the Gnuplot caption on the slide's picture reads $\frac{3xy}{y^2+x^2}$, a different function whose limit at the origin does not exist (along $y = mx$ it is $\frac{3m}{1+m^2}$). Figure G3.1 redraws the function in the text.
3. Slide 001·12 says the function is not continuous at $(0, 0)$ "as $f(0, 0)$ is not defined". Added for clarity: the limit is 0, so defining $f(0, 0) = 0$ removes the discontinuity (a removable discontinuity); with $2xy/(3x^2 + y^2)$ (slide 001·13) no value of $f(0, 0)$ helps.

4. The lecture says a function is continuous "if the surface is smooth" (T001b 00:06:26). Continuity only needs no holes or jumps; a continuous surface can have a sharp edge or corner ($z = |x| + |y|$). Corrected as an exam trap.
5. Slide 001·17 says mixed partials "are in general equal". Precisely, Clairaut's theorem needs f_{xy} and f_{yx} continuous near the point; the standard counter-example $\frac{xy(x^2-y^2)}{x^2+y^2}$ has $f_{xy}(0,0) = -1 \neq f_{yx}(0,0) = 1$ (added for clarity; sympy).
6. The lecture asks the class to prove Clairaut's theorem (T001b 00:20:10) and says the proof is short. The proof needs the mean value theorem as well as the limit definition; it is not on the slides and is not reproduced here.
7. The lecture's history (T001b 00:10:53) says Leibniz used other letters for partial derivatives in a letter to L'Hôpital. The letters are not on the slide and could not be confirmed from the material; only the slide's facts (Leibniz, Legendre 1786, Jacobi 1841) are kept.
8. The lecture's example function (T001a, T001b 00:15:16) adds a term in t that the recognition renders unclearly; W4.2 keeps the clearly spoken terms $x^3 \sin y + e^w + \log p$, which show the same point.
9. The lecture opens with "Unit Two" (T001a 00:00:01); partial differentiation is Unit 1 of UE26MA141A.
10. The deck's motivation slides (001·5–7) carry stock images (a cathode-ray tube, a weather chart, COVID-19 posters); they are described in words, not redrawn.
11. Transcript recognition errors were corrected silently: "Clydoids" → Clairaut's, "legendary" → Legendre, "Lobithal" → L'Hôpital, "Ebo" → above, "parable" → parabola, "cathode ratio" → cathode-ray tube, "dou" / "Do" → ∂, "x q" / "XQ" → x^3, "sine 1" → $\sin y$, "topol o topography" → topography, "extending to" → x tending to.

Coverage

Source	Citations	Slides cited (slide × citations)
S001	56	4×1, 5×1, 6×1, 7×1, 8×4, 9×8, 10×5, 11×8, 12×5, 13×3, 14×2, 15×6, 16×3, 17×8
CW	5	
ESA	1	
T001a	39	
T001b	44	

Slides 001·1, 001·2 and 001·18 are title, session and closing slides. Slides 001·3–7 are motivation and are summarised in topic 1. The lecture announces the geometric interpretation (T001b 00:20:38); it is class 002.